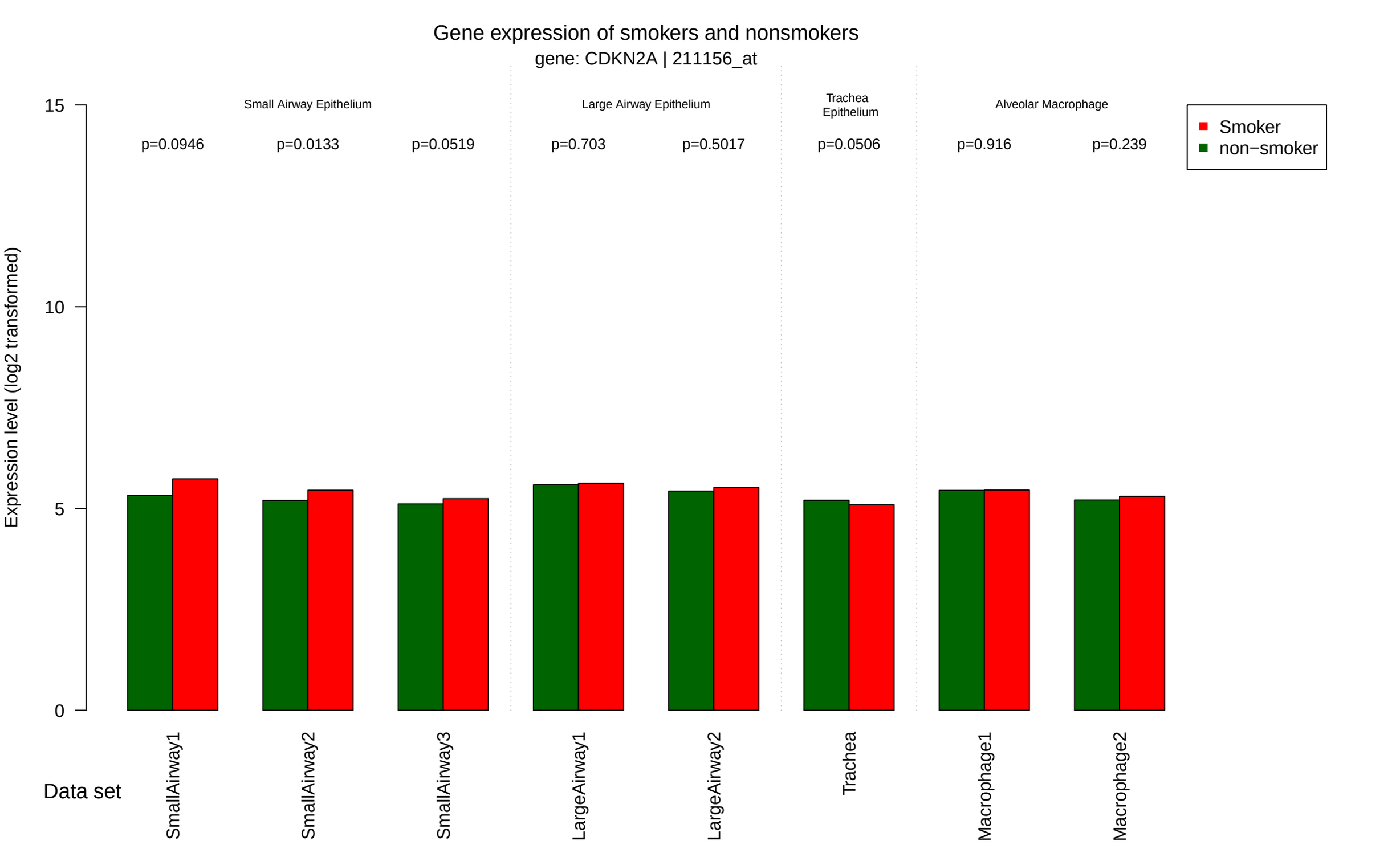
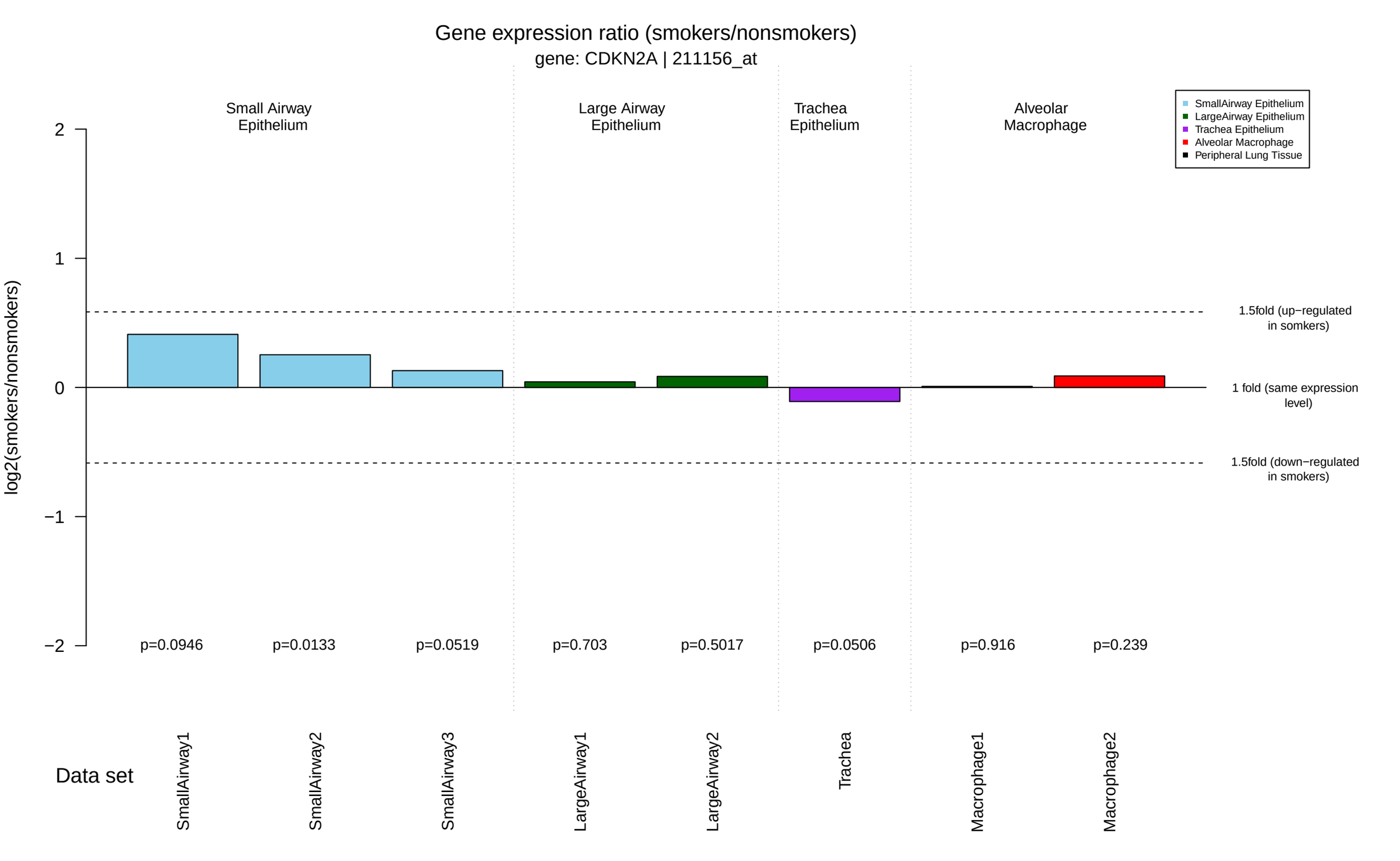
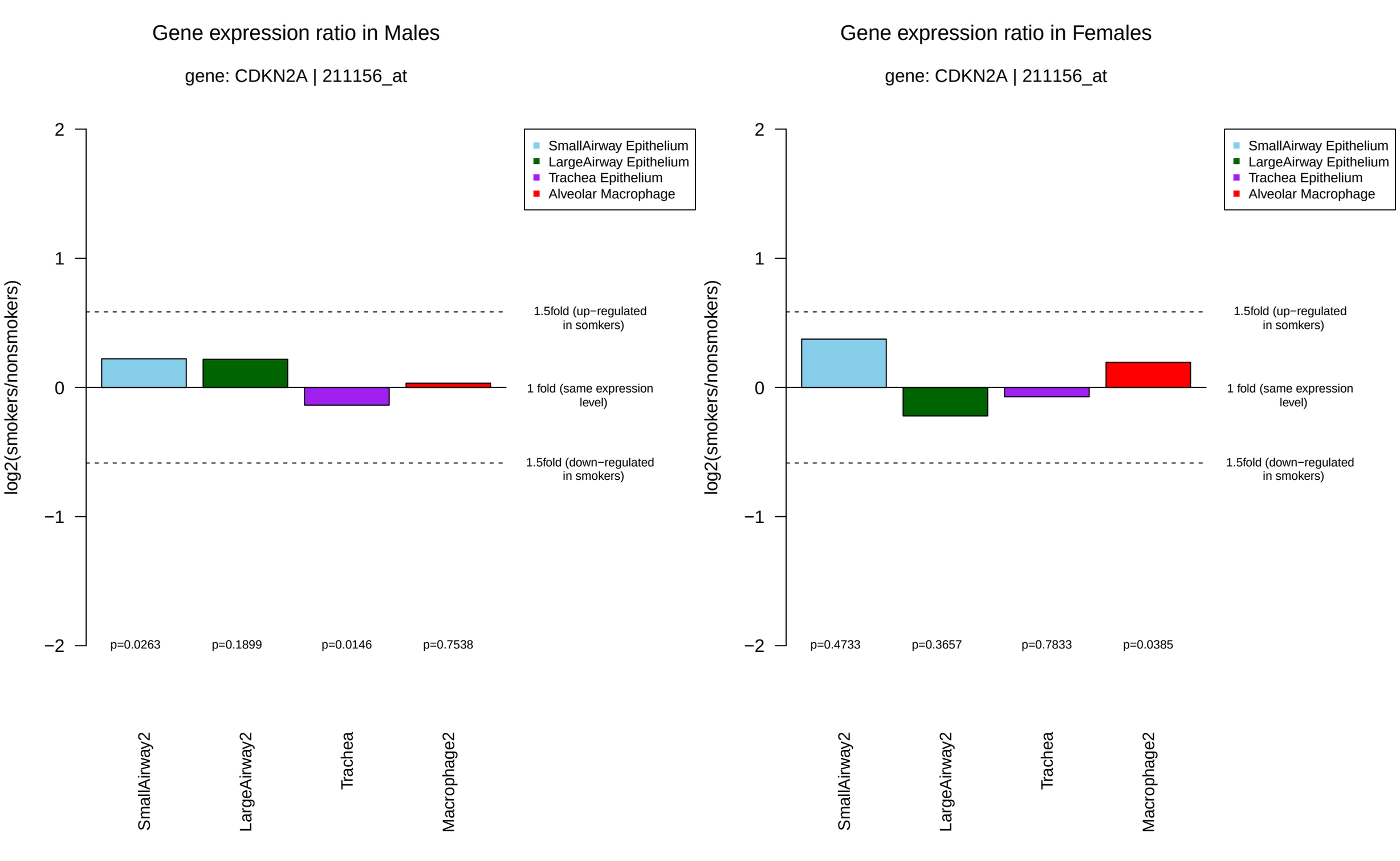
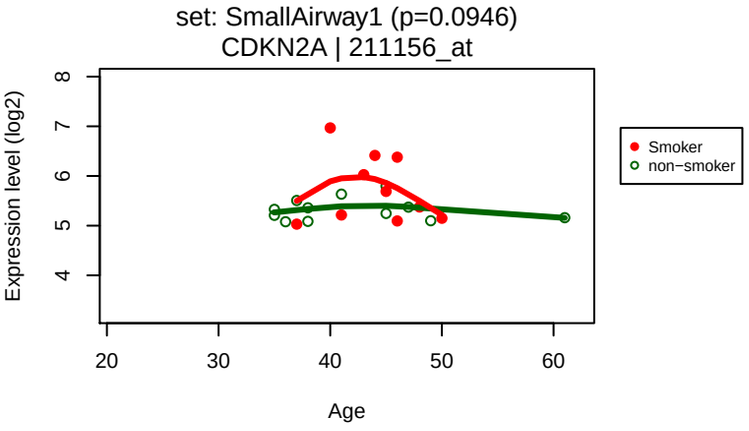
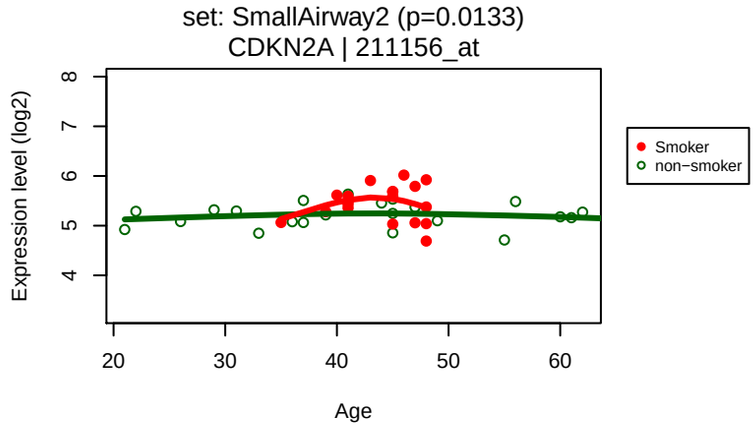
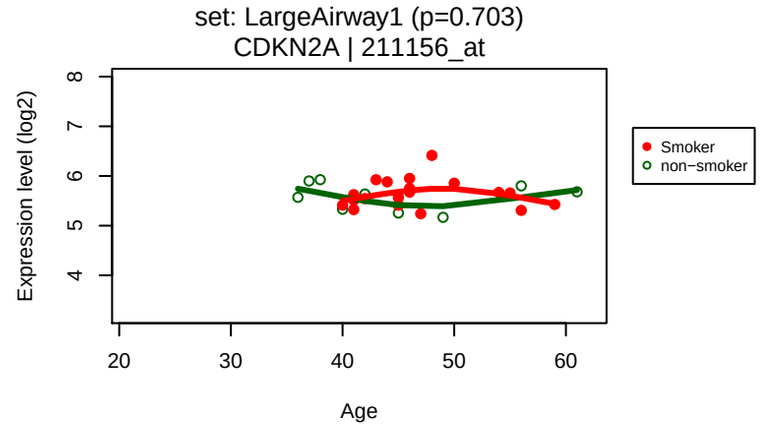

Gene expression of smokers and nonsmokers
gene: CDKN2A | 211156_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0946
p=0.0133
p=0.0519
p=0.703
p=0.5017
p=0.0506
p=0.916
p=0.239
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

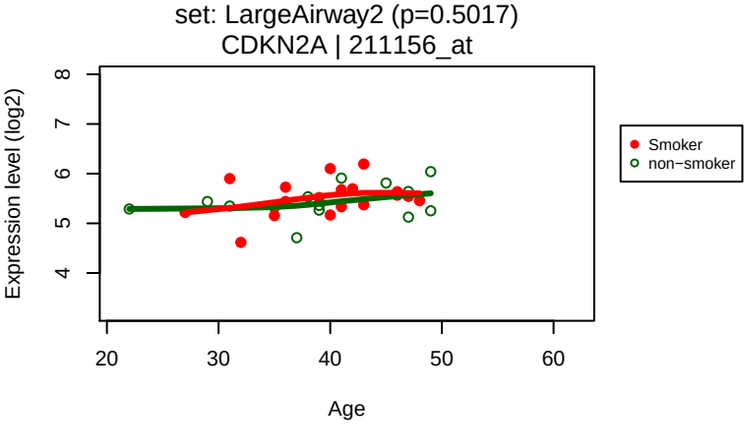

Gene expression ratio (smokers/nonsmokers)
gene: CDKN2A | 211156_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0946
p=0.0133
p=0.0519
p=0.703
p=0.5017
p=0.0506
p=0.916
p=0.239
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

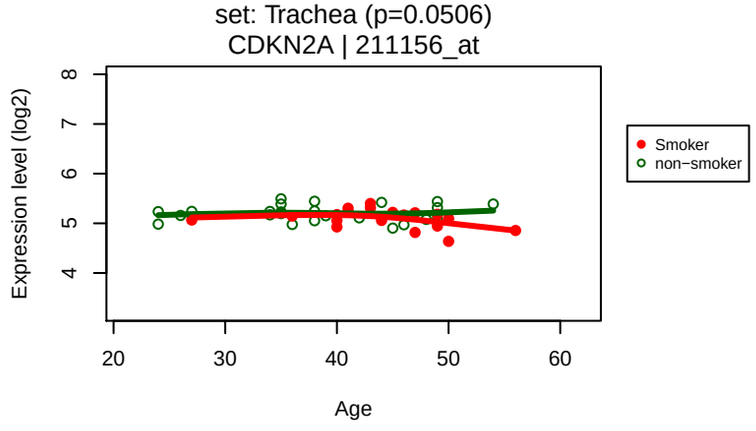

Gene expression ratio in Males
gene: CDKN2A | 211156_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0263
p=0.1899
p=0.0146
p=0.7538
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CDKN2A | 211156_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4733
p=0.3657
p=0.7833
p=0.0385
SmallAirway2
LargeAirway2
Trachea
Macrophage2

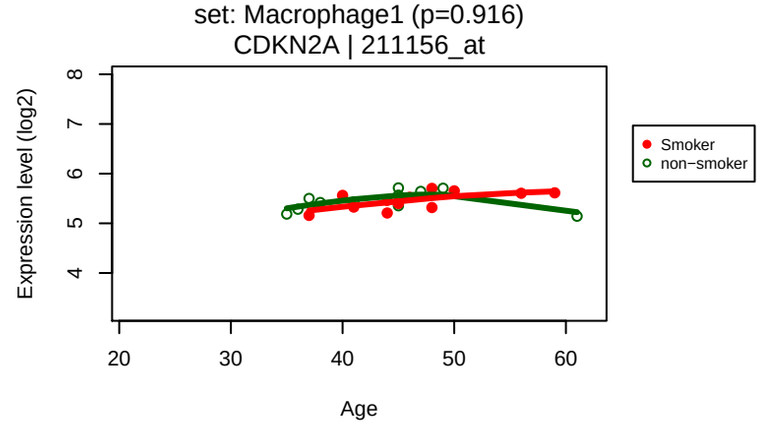

set: SmallAirway1 (p=0.0946)
CDKN2A | 211156_at
Expression level (log2)
Age
Smoker
non−smoker

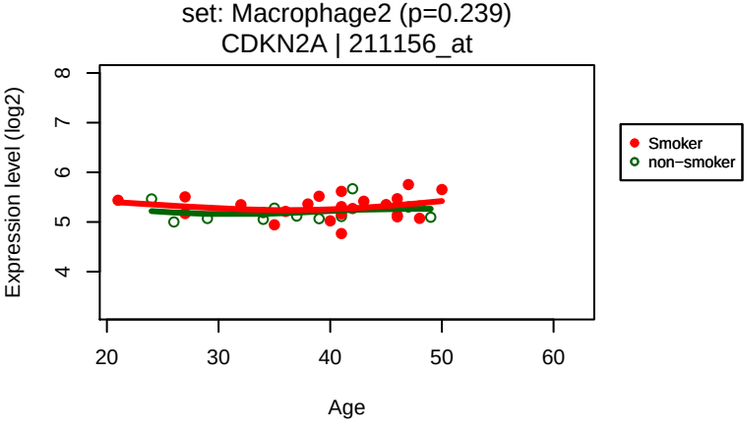

set: SmallAirway2 (p=0.0133)
CDKN2A | 211156_at
Expression level (log2)
Age
Smoker
non−smoker

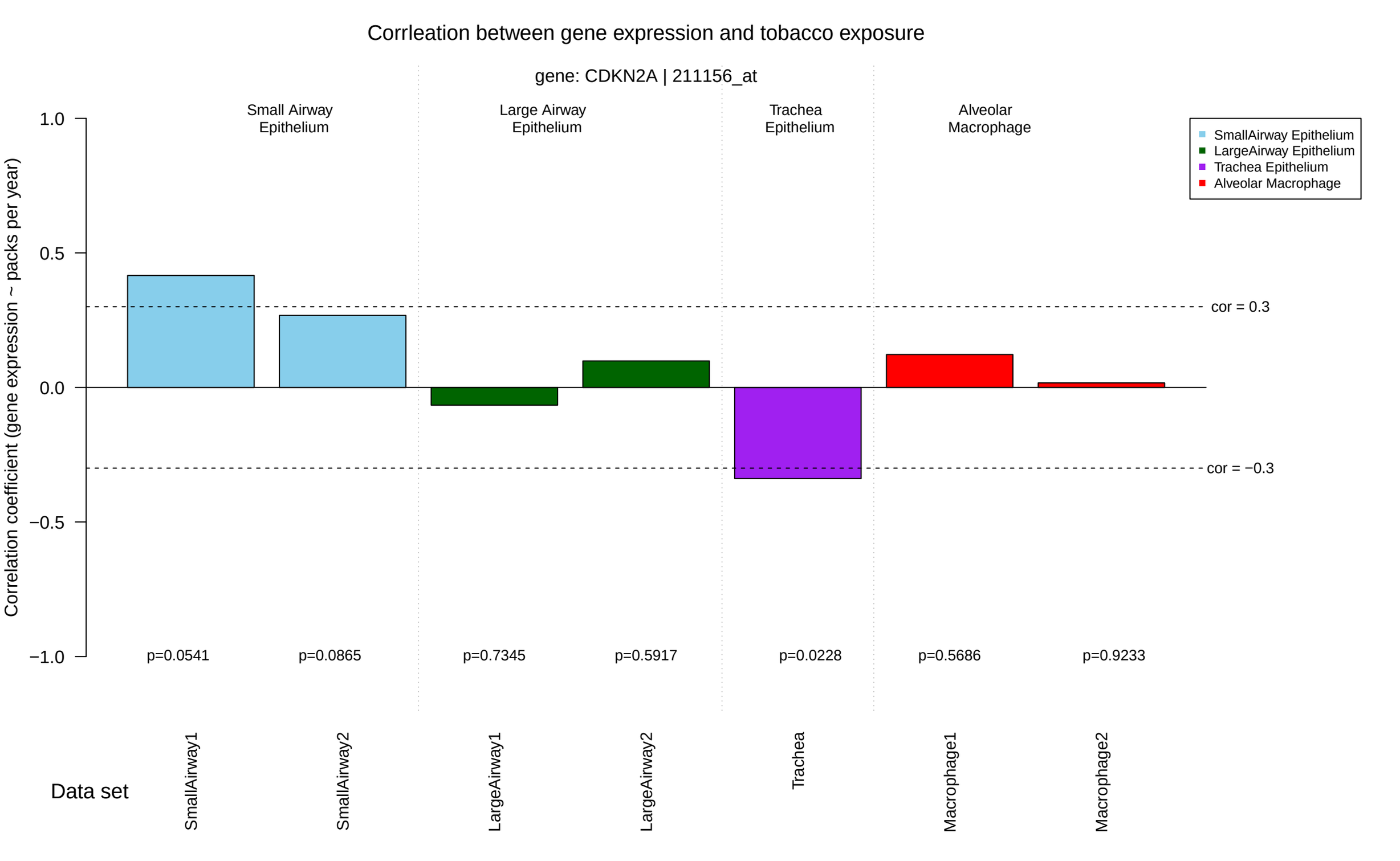

set: LargeAirway1 (p=0.703)
CDKN2A | 211156_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5017)
CDKN2A | 211156_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0506)
CDKN2A | 211156_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.916)
CDKN2A | 211156_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.239)
CDKN2A | 211156_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CDKN2A | 211156_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0541
p=0.0865
p=0.7345
p=0.5917
p=0.0228
p=0.5686
p=0.9233
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2